



SBA 927 - Business Text Analytics: Natural Language Processing Techniques

Version 1.0, 09/13/2024

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Introduction

Welcome to a practical exploration of Natural Language Processing (NLP) techniques applied to business analytics! In this assessment, you will step into the role of a data analyst tasked with enhancing business decision-making through text data analysis.

Imagine that you are employed by a company that aims to enhance its customer experience and strengthen its market position. As a data analyst, your primary responsibility is to analyze various text data sources such as customer reviews, market sentiment reports, and internal business documents.

In this assessment, you will be tasked with identifying key themes and trends from the text data to inform business decisions. You will utilize essential NLP techniques to extract valuable insights such as sentiment analysis to gauge customer satisfaction and topic modeling to uncover prevalent issues or interests.

Furthermore, you will leverage pre-trained language models to refine your analysis, generating actionable insights that can guide the company's strategy. By the end of this assessment, you will have a clear understanding of how to apply NLP techniques in real-world business scenarios, preparing you to drive data-informed strategies and decisions in your future job as a data analyst.

Grading

Criteria	Exemplary (20 pts)	Proficient (10-19 pts)	Needs Improvement (0-5 pts)
Text Data Processing (Tokenization, Stemming, Lemmatization)	Text data is thoroughly processed with accurate tokenization, noise removal, normalization, stop word removal, and effective application of stemming and lemmatization. The results are clean and well-documented.	Most text data is processed correctly, but there may be minor issues in tokenization or preprocessing steps. Stemming and lemmatization are applied, with some results needing refinement.	Significant issues with text processing or incomplete application of tokenization, stemming, and lemmatization. Results may be inconsistent or poorly documented.
Named Entity Recognition and Part-of-Speech Tagging	Accurate and comprehensive identification of named entities and part-of-speech tags. Results are well-analyzed, with clear explanations of how they contribute to understanding the text.	Named entities and POS tags are identified correctly, but there may be some inaccuracies or lack of depth in the analysis. Results are mostly clear.	Inaccurate or incomplete identification of named entities and POS tags. Analysis lacks clarity or depth.
Sentiment Analysis and Insights	Effective creation of prompts for sentiment analysis with accurate identification of sentiment types (positive, negative, neutral). Insights into model performance are detailed and well-supported.	Sentiment analysis prompts are created with generally accurate results. Insights are provided but may lack some depth or clarity.	Inaccurate sentiment analysis prompts or results. Insights into model performance are minimal or unclear.
Quality of Analysis and Reporting	The report is thorough, well-structured, and provides clear insights into prompt generation, fine-tuning, and bias analysis. All deliverables are addressed with high quality and detail.	The report is complete and generally well-structured but may lack some detail or clarity in certain areas. Most deliverables are addressed adequately.	The report is incomplete, lacks detail, or is poorly structured. Several deliverables may be missing or inadequately addressed.

Objectives

This assessment introduces learners to core Natural Language Processing (NLP) techniques through hands-on practice in business-related text analysis, emphasizing skills applicable to real-world scenarios such as customer sentiment analysis and key insight extraction from documents. Learners will gain practical experience with:

- **Applying** fundamental NLP techniques, including tokenization, stemming, lemmatization, and part-of-speech tagging.
- **Performing** sentiment analysis to interpret customer feedback and business text data.
- **Identifying** named entities and extracting key insights from business documents using NLP methods.
- **Experimenting** with pre-trained language models for enhanced text analysis and business insights.

Prerequisites

To complete this assessment:

- For this assessment, learners are expected to have:
- A basic understanding of NLP concepts and access to NLP libraries and tools. You can use any publicly available IDE with Python support such as:
 - **Replit:** <https://replit.com>
 - **Google Colab:** <https://colab.research.google.com>
 - **Jupyter Notebook:** <https://jupyter.org>
- Libraries: NLTK (Natural Language Toolkit), spaCy, Hugging Face Transformers, and any additional libraries as required.
- You will need a dataset to continue with the requirements below. Find the [link](#) here.

Submission

Upload your report to the designated submission link.

Requirements

Text Data Processing:

- **Tokenization and Preprocessing:**

Load the Dataset: Begin by loading the provided dataset. Ensure that you understand the format and content of the data before proceeding.

Clean the Text Data:

- **Remove Noise:** Eliminate irrelevant characters such as special symbols and extra whitespace.
- **Normalize Text:** Convert all text to lowercase to ensure uniformity and avoid treating the same words in different cases as different terms.
- **Remove Punctuation:** Strip out punctuation marks such as commas, periods, and question marks that are not essential for the analysis.

- **Remove Stop Words:** Identify and remove common stop words (e.g., "the," "and," "is") that may not contribute significant meaning to the analysis.

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

import string


# Download NLTK resources if not already present

nltk.download('punkt')

nltk.download('stopwords')


# Function for Tokenization and Preprocessing

def tokenize_and_preprocess(text):

    # Tokenize the text

    tokens = word_tokenize(text)


    # Remove stop words and punctuation

    stop_words = set(stopwords.words('english'))

    punctuation = set(string.punctuation)


    filtered_tokens = [word.lower() for word in tokens if (word.isalpha() and word.lower() not in
stop_words and word not in punctuation)]


    return filtered_tokens


# Read the external file

file_path = 'SBA927.txt'

with open(file_path, 'r', encoding='utf-8') as file:

    # Read the entire content of the file

    dataset = file.read().splitlines()
```

```
# Tokenization and Preprocessing for the entire dataset
```

```
processed_dataset = [tokenize_and_preprocess(text) for text in dataset]
```

```
# Display the results
```

```
for i, text_tokens in enumerate(processed_dataset):
```

```
    print(f"Original Text {i + 1}: {dataset[i]}")
```

```
    print(f"Processed Tokens {i + 1}: {text_tokens}")
```

```
    print("\n")
```

- **Stemming and Lemmatization:**

Apply stemming and lemmatization techniques to the preprocessed text data. This will help in normalizing words and identifying root forms, essential for tasks like sentiment analysis or keyword extraction in business applications.

Import necessary libraries

import nltk

from nltk.stem import PorterStemmer, WordNetLemmatizer

Download NLTK resources if not already present

nltk.download('wordnet')

Define a function for stemming and lemmatization

def stem_and_lemmatize(tokens):

Initialize stemming and lemmatization tools

stemmer = PorterStemmer()

lemmatizer = WordNetLemmatizer()

Apply stemming

stemmed_tokens = [stemmer.stem(token) for token in tokens]

Apply lemmatization

lemmatized_tokens = [lemmatizer.lemmatize(token) for token in tokens]

return stemmed_tokens, lemmatized_tokens

Assume 'processed_dataset' contains the preprocessed text data

Apply stemming and lemmatization to the entire dataset

stemmed_and_lemmatized_dataset = [stem_and_lemmatize(tokens) for tokens in processed_dataset]

Display the results

for i, (stemmed_tokens, lemmatized_tokens) in enumerate(stemmed_and_lemmatized_dataset):

print(f"Original Tokens {i + 1}: {processed_dataset[i]}")

print(f"Stemmed Tokens {i + 1}: {stemmed_tokens}")

```
print(f"Lemmatized Tokens {i + 1}: {lemmatized_tokens}")  
  
print("\n")
```

Named Entity Recognition and Part-of-Speech Tagging:

- **Named Entity Recognition (NER):**

Analyze sample sentences and use NLP libraries to identify and categorize named entities in the text.

Import necessary libraries

import nltk

from nltk.tokenize import sent_tokenize, word_tokenize

from nltk import pos_tag, ne_chunk

Download required resources

nltk.download('punkt')

nltk.download('averaged_perceptron_tagger')

nltk.download('maxent_ne_chunker')

nltk.download('words')

Function to tokenize text into sentences

def tokenize_into_sentences(text):

sentences = sent_tokenize(text)

return sentences

Function to perform part-of-speech tagging on sentences

def part_of_speech_tagging(sentences):

pos_tagged_sentences = [pos_tag(word_tokenize(sentence)) for sentence in sentences]

return pos_tagged_sentences

Function to extract named entities from POS-tagged sentences

def extract_named_entities(pos_tagged_sentences):

named_entities = []

for sentence in pos_tagged_sentences:

tree = ne_chunk(sentence)

for subtree in tree:

if hasattr(subtree, 'label'): # Check if subtree is a named entity

```
entity = " ".join([word for word, tag in subtree.leaves()])
```

```
named_entities.append((entity, subtree.label()))
```

```
return named_entities
```

```
# Define a function for Named Entity Recognition
```

```
def named_entity_recognition(text):
```

```
    # Tokenize and process the text
```

```
    sentences = tokenize_into_sentences(text)
```

```
    pos_tags = part_of_speech_tagging(sentences)
```

```
    named_entities = extract_named_entities(pos_tags)
```

```
    return named_entities
```

```
# Apply Named Entity Recognition to the entire dataset
```

```
ner_results_dataset = [named_entity_recognition(text) for text in dataset]
```

```
# Display the results
```

```
for i, ner_results in enumerate(ner_results_dataset):
```

```
    print(f"Named Entities in Text {i + 1}: {ner_results}")
```

```
    print("\n")
```

- **Part-of-Speech (POS) Tagging:**

Apply POS tagging on the text data to label and categorize the parts of speech in business documents, such as reports or emails.

```
# Import necessary libraries

import nltk

from nltk import pos_tag

from nltk.tokenize import word_tokenize, sent_tokenize


# Download NLTK resources if not already present

nltk.download('punkt')


# Assume 'processed_dataset' contains the preprocessed text data


# Define a function for POS tagging

def pos_tagging(text):

    # Tokenize and process the text

    sentences = sent_tokenize(text)

    pos_tags = []

    # Process each sentence

    for sentence in sentences:

        words = word_tokenize(sentence)

        pos_tags.extend(pos_tag(words))

    return pos_tags


# Apply POS tagging to the entire dataset

pos_tags_dataset = [pos_tagging(text) for text in dataset]


# Display the results

for i, pos_tags in enumerate(pos_tags_dataset):

    print(f"POS Tags in Text {i + 1}: {pos_tags}")

    print("\n")
```

Sentiment Analysis:

- **Prompt Creation for Sentiment Analysis:**

Create prompts to perform sentiment analysis on business-related text such as customer feedback or product reviews. Identify whether the sentiment is positive, negative, or neutral, and provide insights into the accuracy and effectiveness of the model's response.

- **Observation and Analysis:**

Use the prompts created in Requirement 5 to perform sentiment analysis on sample sentences. Evaluate the responses generated by the language model and provide insights into their accuracy and effectiveness.

Pre-trained Language Models:

- **Experiment with Pre-trained Models:**

Experiment with pre-trained language models to analyze business text. Use models for tasks like tokenization, NER, POS tagging, and sentiment analysis, and evaluate how they perform in a business context.

Time/Duration

This assessment will be conducted over one day, dedicated to different NLP techniques applied to business text.

Deliverables

Submit a report that includes:

- Processed text data with tokenization and preprocessing (Requirement 1 and 2).
- Stemmed and lemmatized text data (Requirement 2).
- Results of Named Entity Recognition and Part-of-Speech Tagging (Requirements 3 and 4).
- Prompts for sentiment analysis (Requirement 5).
- Sentiment analysis results and insights (Requirement 6).
- Observations from experimenting with pre-trained language models (Requirement 7).
- Please use this [perscholas template](#) for discussing the differences and advantages of each technique.